EDITORIAL

Preventing Delirium in the Intensive Care Unit

Anthony Delaney, MBBS, MSc, PhD; Naomi Hammond, MN, MPH, PhD; Edward Litton, MBChB, MSc, PhD

Delirium is a complex problem, has a poorly understood pathophysiology, and usually manifests as a fluctuating disturbance of consciousness and cognition that develops over a short period and is associated with physical or mental illness.[1] Delirium is not exclusively a complication of critical illness but develops in approximately one-third of patients treated in intensive care units (ICUs).[2] The majority of these patients experience hypoactive delirium, a quietly confused state, rather than the more notable cases of agitated delirium, which present as overt hallucinations and hyperactivity.[3]

Related article page 680

Delirium is costly to the health system and, importantly, to patients. Patients who experience an episode of delirium are at an increased risk of prolonged mechanical ventilation, prolonged ICU length of stay, and increased hospital mortality[2] and are more likely to develop long-term cognitive impairment.[4] Irrespective of whether the link with adverse outcomes is causal, delirium is a distressing symptom for patients and their families. Visual and auditory hallucinations and impairment of cognition render delirious patients unable to determine reality from the multiple perceived threats in their environment. As a result, survivors of critical illness frequently report being more traumatized by memories of hallucinations than of real events that occurred during their ICU stay.[5]

Although the risk factors for delirium and its natural history are well described, evidence that pharmacological therapy may be effective for preventing delirium is lacking. Antipsychotic medications, often used to manage the symptoms of established delirium, represent plausible candidate interventions for pharmacological prophylaxis. Thus, high-quality trials investigating strategies to reduce the incidence, duration, and adverse sequelae following an episode of delirium address an important unmet need.

In this issue of *JAMA*, the findings from the REDUCE trial, reported by van den Boogaard and colleagues,[6] provide substantial new insights. In this multicenter, placebo-controlled, randomized clinical trial conducted in 21 centers in the Netherlands, the investigators compared the effect of haloperidol to placebo among patients at high risk of developing delirium. The study was conducted according to a prepublished protocol, with an adaptive design to allow efficient use of information gathered as the trial progressed. The primary outcome, number of days of survival over 28 days, was robust, patient-centered, and reasonable given prior observational studies. Importantly, the REDUCE Study Investigators sought to ensure that participating sites had in place strategies for reducing delirium prior to commencing enrollment. These strategies included early mobilization, improving the circadian rhythm of the trial participants, noise reduction, sedation protocols aimed to prevent oversedation, reducing the use of benzodiazepines, and the use of hearing and visual aids. By standardizing "best practice" as routine care, the researchers aimed to determine whether the addition of a pharmacological therapy improved survival in a cohort of patients at high risk of developing delirium.

At 28 days, there was no significant improvement in the primary outcome of number of days of survival among patients allocated to receive treatment (haloperidol 2 mg, 3 times a day) compared with those allocated to receive matching placebo (median, 28 days for both groups; difference, 0; 95%CI, 0-0), and there was no significant difference in the proportion of patients surviving by the 28th day, with survival rates of 610 of 732 patients (83.3%) in the treatment group vs 585 of 707 patients (82.7%) in the placebo group (absolute difference, 0.6%; 95% CI, −3.4% to 4.6%). In addition, there was no effect of prophylactic haloperidol on reducing the incidence of delirium. Delirium was diagnosed in 244 of 732 trial participants (33.3%) in the treatment group and in 233 of 707 trial participants (33.0%) in the placebo group, a baseline in keeping with the expected incidence of delirium in a critically ill population.[2]

There are some questions regarding the ascertainment of delirium as a crucial secondary end point. This outcome was measured using 2 different instruments, dependent on site: the number of delirium-free and coma-free days was collected in only two-thirds of participating sites, daily assessment for 28-day delirium- and coma-free days was only conducted in the ICU, and patients without delirium at ICU discharge were considered delirium free. These shortcomings, however, are unlikely to hide a substantial treatment effect because there was no effect seen in any of the other secondary outcomes. Both groups had similar duration of mechanical ventilation, ICU and hospital length of stay, rates of unplanned removal of tube and catheters, and rates of reintubation. In short, haloperidol had no demonstrable effect on any outcome measured in the trial.

In the most narrow sense, these are generalizable results that offer no rationale for the continued use of haloperidol as a prophylactic measure in critically ill patients to improve survival, reduce the incidence of delirium, or decrease the harms associated with delirium. In a broader sense, although the trial failed to demonstrate superiority, it would not be appropriate to label the findings as negative, considering the wealth of valuable and new information provided by the REDUCE investigators and trial participants.

Several lessons can be taken from the results of this study. First, the investigators have shown that patients at

© 2018 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by University of Chicago Libraries user on 01/28/2019

high risk of developing delirium can be identified using simple means. Participants were included in the trial solely based on whether their physician anticipated that their patients' ICU length of stay would be more than 2 days. Approximately one-third of all trial participants who met this single criterion developed measurable delirium. Thus, this very simple strategy was highly effective at selecting a population at high risk of delirium, potentially obviating the need for more complex screening tools when recruiting patients in future trials.

Second, the detailed reporting of the interventions provided as part of routine care aimed at reducing the incidence of delirium is enlightening. The failure of the results of single-center studies to be replicated in multicenter trials[7] may in part be due to the heterogeneity of background treatments and care across multiple clinical sites. Detailed reporting of usual care procedures, relevant to the question at hand, assists clinicians in deciding how applicable the results of a particular clinical trial are to their practice. This type of reporting also assists researchers in building on the methods in subsequent studies so that the heterogeneity of background noise can be minimized.

Third, in demonstrating that haloperidol was not effective in preventing delirium or reducing the harms associated with delirium, the REDUCE study challenges the current model that the addition of psychoactive medication to patients who are already receiving multiple interventions may be beneficial. As further research efforts are undertaken to identify comparatively less harmful medications to prevent delirium,[8] the results should be carefully interpreted in light of the effect that such a treatment has in addition to the provision of the suite of standard-care nonpharmacological interventions. It may be that no single pharmacological intervention can provide a solution to a condition that has multiple risk factors, occurs in patients receiving complex and invasive therapy, and is associated with significant comorbidity and severely deranged acute physiology.[9]

Compared with pharmacological prophylaxis, an attractive property of nonpharmacological interventions to prevent delirium is that these approaches generally involve either doing less for patients (avoiding excessive sedation, benzodiazepines, nocturnal noise, and stimulation) or ensuring the continued provision of relatively simple therapies (mobilization, maintaining a day-night schedule and noise reduction), irrespective of the complexity and challenge of the environment. Although some of these interventions may require planning and cooperation of a multidisciplinary team,[10] a strength of ICU care in general, other interventions may be as simple as providing earplugs in an attempt to reduce the perception of noise and improve sleep.[11] Although the REDUCE study investigators sought to implement best practice, definitive studies supporting the use of nonpharmacological therapies are generally lacking. Future research directed at improving the evidence base of simple interventions such as sleep hygiene bundles (eg, earplugs, eye patches, music therapy, and reduced noise levels) is warranted.

Delirium is a common accompaniment of critical illness, but it need not be so. The REDUCE study has demonstrated that in critically ill patients currently receiving best-practice nonpharmacological interventions to prevent delirium, the addition of haloperidol does not improve survival nor reduce the incidence of delirium or the harms associated with delirium. The REDUCE study has demonstrated that the plausible and simple administration of prophylactic haloperidol is not the solution for the complex problem of delirium in critically ill patients.

ARTICLE INFORMATION

Author Affiliations: Malcolm Fisher Department of Intensive Care Medicine, Royal North Shore Hospital, St Leonards, New South Wales, Australia (Delaney, Hammond); Northern Clinical School, Sydney Medical School, University of Sydney, St Leonards, New South Wales, Australia (Delaney); ANZICS Research Centre, School of Public Health and Preventative Medicine, Monash University, Melbourne, Victoria, Australia (Delaney); Critical Care and Trauma Division, George Institute for Global Health, Newtown, New South Wales, Australia (Hammond); Faculty of Medicine, University of New South Wales, Sydney, Kensington, Australia (Hammond); Fiona Stanley Hospital, Perth, Western Australia (Litton); St John of God Hospital Subiaco, Perth, Western Australia (Litton).

Corresponding Author: Anthony Delaney, MBBS, MSc, PhD, Malcolm Fisher Department of Intensive Care Medicine, Royal North Shore Hospital, Level 6, Acute Services Bldg, Reserve Road, St Leonards, NSW, 2065 Australia (adelaney@med.usyd.edu.au).

Conflict of Interest Disclosures: All authors have completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest and none were reported.

REFERENCES

1. Reade MC, Finfer S. Sedation and delirium in the intensive care unit. *N Engl J Med*. 2014;370(5):444-454.

2. Salluh JI, Wang H, Schneider EB, et al. Outcome of delirium in critically ill patients: systematic review and meta-analysis. *BMJ*. 2015;350:h2538.

3. Peterson JF, Pun BT, Dittus RS, et al. Delirium and its motoric subtypes: a study of 614 critically ill patients. *J Am Geriatr Soc*. 2006;54(3):479-484.

4. Pandharipande PP, Girard TD, Jackson JC, et al; BRAIN-ICU Study Investigators. Long-term cognitive impairment after critical illness. *N Engl J Med*. 2013;369(14):1306-1316.

5. Wade DM, Brewin CR, Howell DC, White E, Mythen MG, Weinman JA. Intrusive memories of hallucinations and delusions in traumatized intensive care patients: an interview study. *Br J Health Psychol*. 2015;20(3):613-631.

6. van den Boogaard M, Slooter AJC, Brüggemann RJM, et al; the REDUCE Study Investigators. Effect of haloperidol on survival among critically ill adults with a high risk of delirium: the REDUCE randomized clinical trial [published February 20, 2018]. *JAMA*. doi:10.1001/jama.2018.0160

7. Bellomo R, Warrillow SJ, Reade MC. Why we should be wary of single-center trials. *Crit Care Med*. 2009;37(12):3114-3119.

8. Shehabi Y, Forbes AB, Arabi Y, et al; The SPICE III study investigators; The Australian and New Zealand Intensive Care Society Clinical Trials Group; The Australian and New Zealand Intensive Care Research Centre. The SPICE III study protocol and analysis plan: a randomised trial of early goaldirected sedation compared with standard care in mechanically ventilated patients. *Crit Care Resusc*. 2017;19(4):318-326.

9. Zaal IJ, Devlin JW, Peelen LM, Slooter AJ. A systematic review of risk factors for delirium in the ICU. *Crit Care Med*. 2015;43(1):40-47.

10. Hodgson CL, Bailey M, Bellomo R, et al; Trial of Early Activity and Mobilization Study Investigators. A binational multicenter pilot feasibility randomized controlled trial of early goal-directed mobilization in the ICU. *Crit Care Med*. 2016;44(6):1145-1152.

11. Litton E, Carnegie V, Elliott R, Webb SA. The efficacy of earplugs as a sleep hygiene strategy for reducing delirium in the ICU: a systematic review and meta-analysis. *Crit Care Med*. 2016;44(5):992-999.

© 2018 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by University of Chicago Libraries user on 01/28/2019